

Classical Statistics for Scientific Data Analysis

Maximum Likelihood Estimates (MLEs)

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Recap: Likelihood

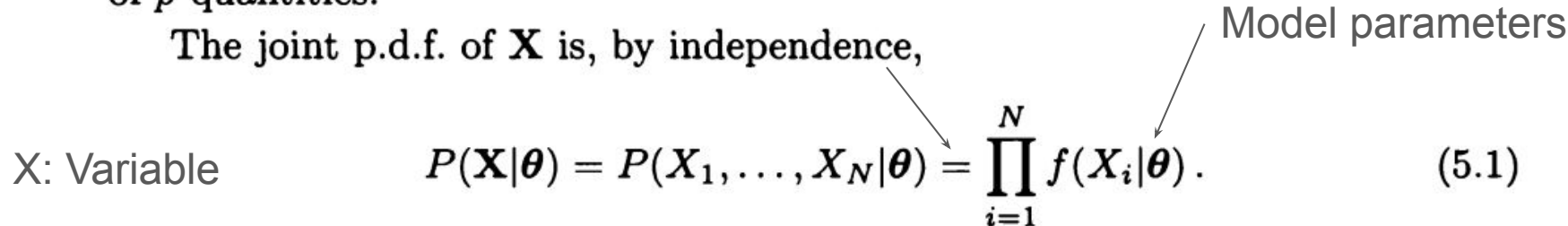
Consider a set of N independent observations of X , say $X_1, \dots, X_N$. These could be N events found in an experiment, an event being the measurement of p quantities.

The joint p.d.f. of $\mathbf{X}$ is, by independence,

X: Variable

$$P(\mathbf{X}|\boldsymbol{\theta}) = P(X_1, \dots, X_N|\boldsymbol{\theta}) = \prod_{i=1}^N f(X_i|\boldsymbol{\theta}). \quad (5.1)$$

Model parameters



When the variable $\mathbf{X}$ is replaced by the observed data $\mathbf{X}^0$, then P is no longer a p.d.f., and it is usual to denote it by L and call $L(\mathbf{X}^0|\boldsymbol{\theta})$ the *likelihood function*, which is now a function of $\boldsymbol{\theta}$ only:

X_0 : Observed data

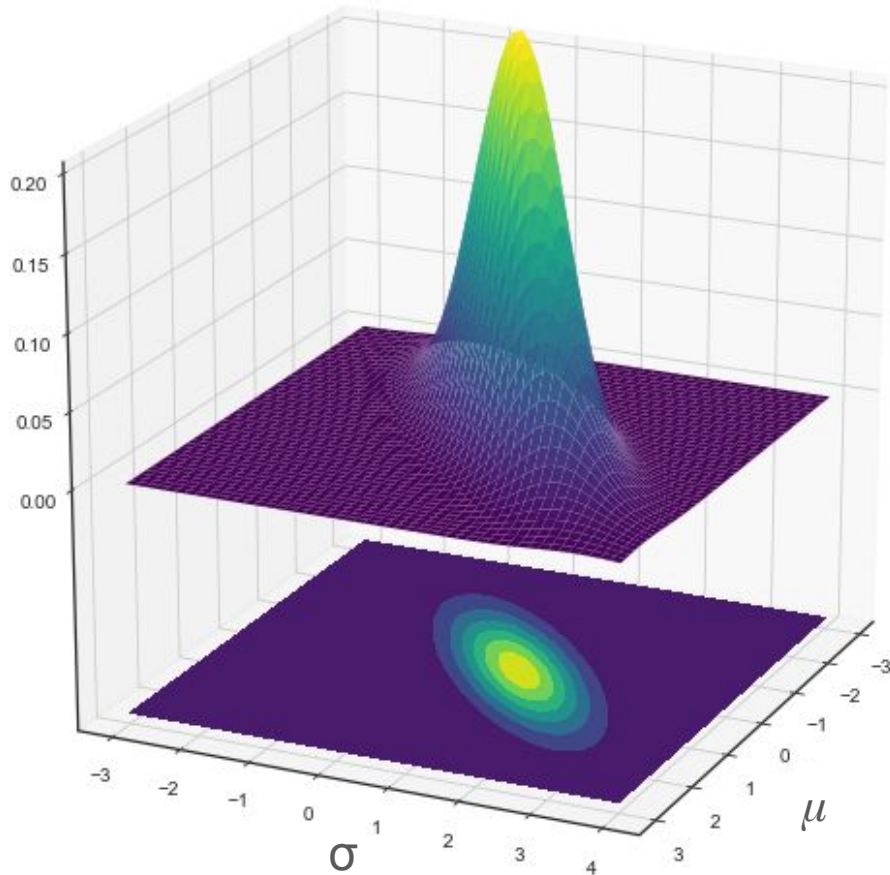
$$L(\boldsymbol{\theta}) = P(\mathbf{X}^0|\boldsymbol{\theta}).$$

Recap: Likelihood example

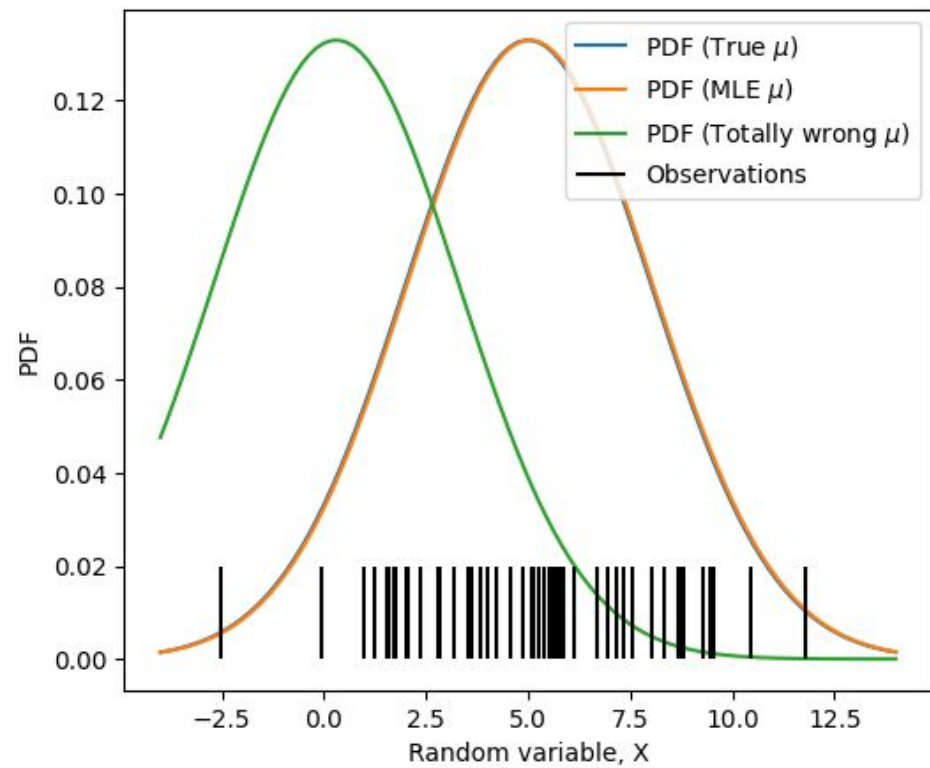
$$\begin{aligned}\mathbf{x} &= \{x_1, x_2, \dots, x_n\} \\ \mathcal{L}(\theta) &= P(\mathbf{x}|\theta) \\ &= \prod_{i=1}^N P(x_i|\theta) \\ &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)\end{aligned}$$

1 number per (μ, σ)

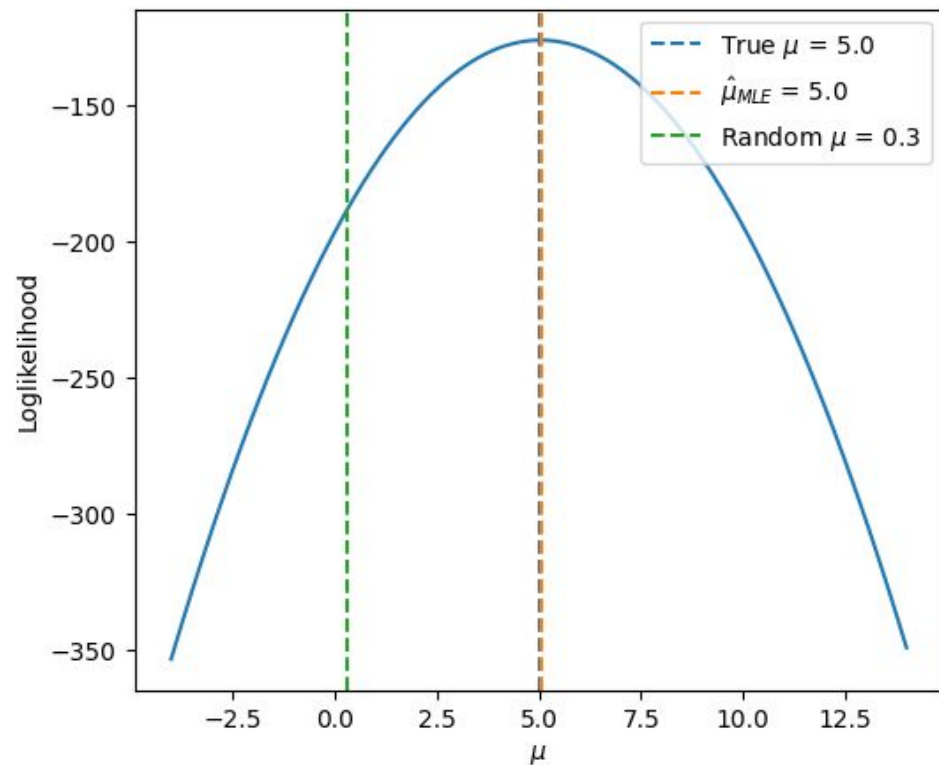
Likelihood surface/function



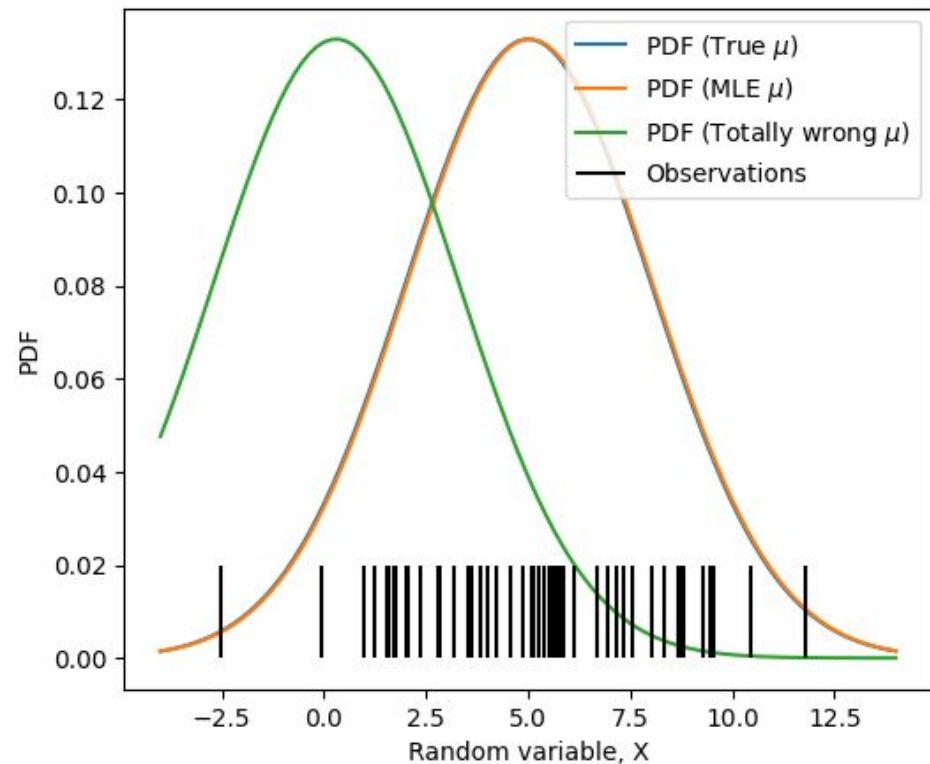
Likelihood intuition



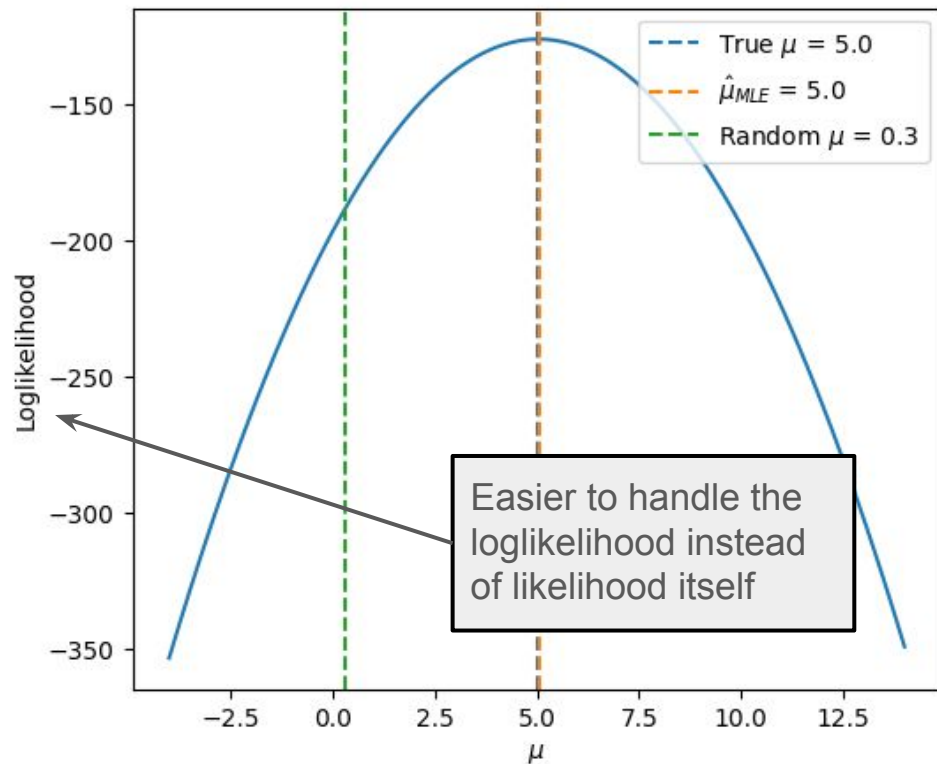
Fixing σ at true value



Likelihood intuition



Fixing σ at true value



Likelihood equation

$$\frac{\partial}{\partial \theta} \sum_{i=1}^N \ln f(X_i, \theta) = \frac{\partial}{\partial \theta} \ln L(\mathbf{X}, \theta) = 0.$$

- Maximum Likelihood Estimate (MLE) is a root of above equation
- Assumes
 - Likelihood differentiable
 - Maximum not at boundaries of parameter
- If more than 1 maximum, take higher one

Recap: Desirable estimator properties

- Consistent

$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta| > \epsilon) = 0.$$

- Unbiased

$$E[\hat{\theta}] - \theta = 0$$

- Efficient

$$V[\hat{\theta}] = \left(1 + \frac{\partial b}{\partial \theta}\right)^2 \bigg/ E \left[-\frac{\partial^2 \log L}{\partial \theta^2} \right],$$

- Robust

- Against outliers, misspecification

Maximum Likelihood Estimator properties

- Consistent

$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta| > \epsilon) = 0.$$

See orangebook Section 7.2.3 or [blog](#) for proof

- Unbiased

$$\lim_{n \rightarrow \infty} E[\hat{\theta}] - \theta = 0$$

Asymptotically unbiased.
See orangebook Eq 7.33.

- Efficient

$$\lim_{n \rightarrow \infty} V[\hat{\theta}] = -\frac{1}{E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]}$$

Asymptotically efficient.
See orangebook Section 7.4.5 for proof.

- Robust

- Against outliers, misspecification

Nope);
Clean up your data first
Misspecification studies

Random comments on MLE

- Nice properties
 1. Consistent estimator
 2. Asymptotically unbiased
 3. Asymptotically efficient
- Irony
 - Last two properties hold for large sample size, $N \rightarrow \infty$
 - But likelihood becomes difficult to compute at large N
- Machine Learning et al.
 - Simulation-based inference (SBI), Likelihood-free inference (LFI), Implicit likelihood inference (ILI)
 - Python packages: [sbi](#), [pydelfi](#), [lampe](#)
 - Train neural network to compute likelihood (or rather, likelihood ratio)
 - E.g. [Learning likelihood ratios with neural network classifiers](#), [Robust Simulation Based Inference](#), many more

MLE example: Exponential distribution

Backstory: muon decay

- Many particles are unstable
 - Decay into other particles

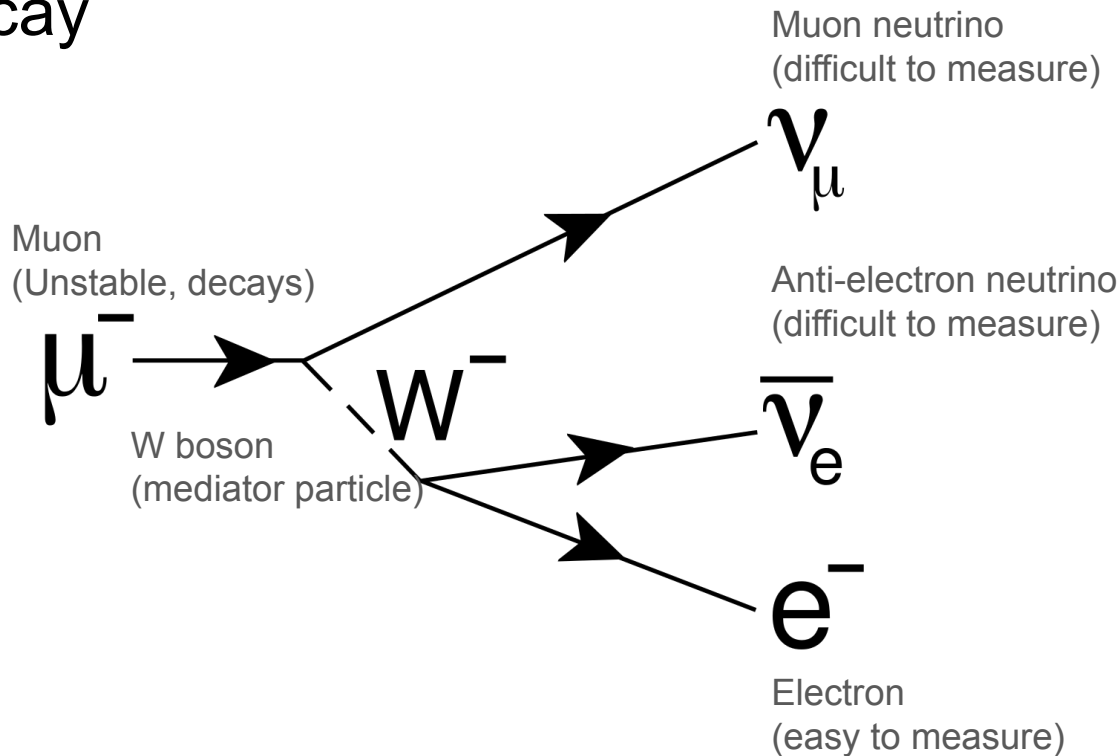
- Probability of decay

$$f_{decay}(t) = Ae^{-\lambda t}$$

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \approx \frac{T_{1/2}}{0.693}$$

Decay constant

- Goal
 - Estimate half-life of muon



Learn about particle decays [here](#)

Likelihood for exponential distribution

- Consider the exponential distribution

- E.g.: Decay time of muons
- τ would be related to half-life

$$f(t; \tau) = \frac{1}{\tau} e^{-t/\tau}$$

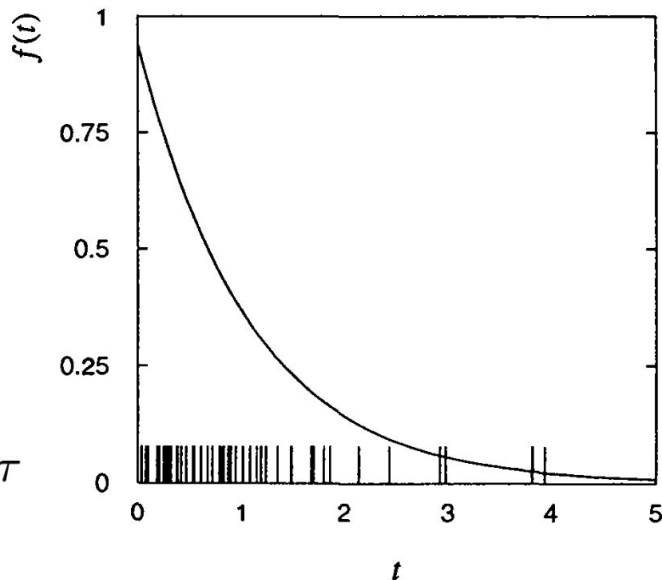
- Suppose we have i.i.d. Data $t_1, \dots, t_n$

- Likelihood function

$$L(\tau) = \prod_{i=1}^n \frac{1}{\tau} e^{-t_i/\tau}$$

- Loglikelihood function

$$\ln L(\tau) = \sum_{i=1}^n \ln f(t_i; \tau) = \sum_{i=1}^n \left(\ln \frac{1}{\tau} - \frac{t_i}{\tau} \right)$$



MLE for exponential distribution

- Find τ that maximises loglikelihood

$$\frac{\partial \ln L}{\partial \tau} = \sum_{i=1}^n \left(-\frac{1}{\tau} + \frac{t_i}{\tau^2} \right) = 0$$

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n t_i$$

Happens to be sample mean

Expectation and Variance of MLE

- Mean and variance for exponential distribution

$$E[t] = \int_0^{\infty} t \frac{1}{\tau} e^{-t/\tau} dt = \tau$$

$$V[t] = \int_0^{\infty} (t - \tau)^2 \frac{1}{\tau} e^{-t/\tau} dt = \tau^2$$

- Given MLE

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n t_i$$

- Expectation of MLE

$$E[\hat{\tau}] = E \left[\frac{1}{n} \sum_{i=1}^n t_i \right] = \frac{1}{n} \sum_{i=1}^n E[t_i] = \tau$$

Unbiased

- Variance of MLE

$$V[\hat{\tau}] = V \left[\frac{1}{n} \sum_{i=1}^n t_i \right] = \frac{1}{n^2} \sum_{i=1}^n V[t_i] = \frac{\tau^2}{n}$$

$$\sigma_{\hat{\tau}} = \frac{\tau}{\sqrt{n}}$$

But recall..

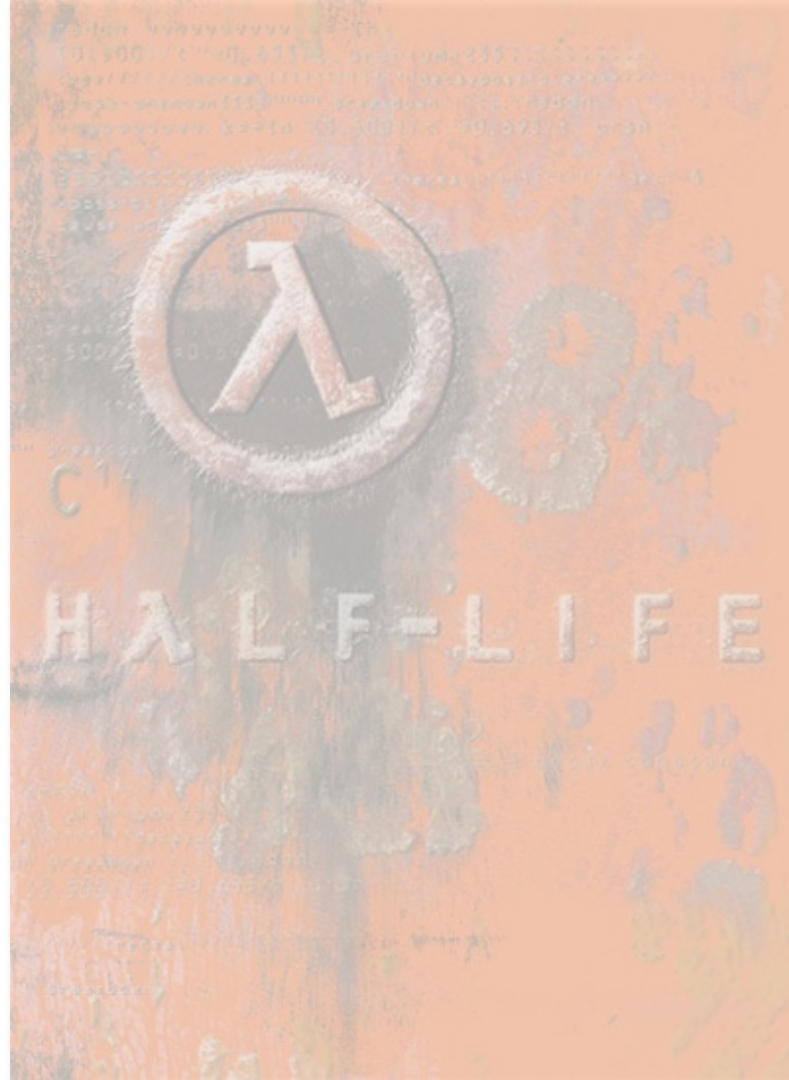
$$f_{decay}(t) = Ae^{-\lambda t}$$

Half-life

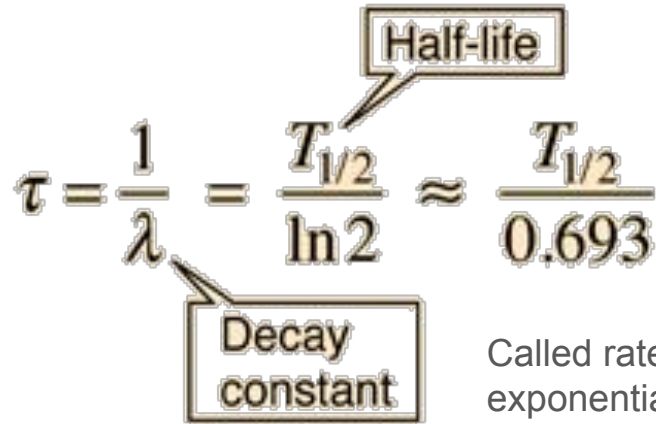
$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \approx \frac{T_{1/2}}{0.693}$$

Decay
constant

Just found
MLE for this



But what happens if you want to formulate the problem in terms of the decay constant, λ ?

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \approx \frac{T_{1/2}}{0.693}$$


The diagram illustrates the relationship between the decay constant and the half-life. It features the equation $\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \approx \frac{T_{1/2}}{0.693}$. A callout box labeled "Half-life" points to the $T_{1/2}$ term in the equation. Another callout box labeled "Decay constant" points to the λ term in the equation.

Called rate parameter in context of exponential distribution

Break

MLE for λ instead after break

Back 10:55 am.

MLE for exponential rate parameter

1. Given exponential distribution $f(t; \lambda) = \lambda e^{-\lambda t}$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data

$$f(t; \lambda) = \lambda e^{-\lambda t}$$
$$\{t_1, t_2, \dots, t_n\}$$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data
3. Write down likelihood function

$$f(t; \lambda) = \lambda e^{-\lambda t}$$

$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data
3. Write down likelihood function
4. Take log

$$f(t; \lambda) = \lambda e^{-\lambda t}$$

$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

$$\ln \mathcal{L}(\lambda) = \sum_{i=1}^n \ln (\lambda e^{-\lambda t_i}) = \sum_{i=1}^n (\ln \lambda - \lambda t_i)$$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data
3. Write down likelihood function
4. Take log
5. Differentiate w.r.t. parameter of interest

$$f(t; \lambda) = \lambda e^{-\lambda t}$$

$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

$$\ln \mathcal{L}(\lambda) = \sum_{i=1}^n \ln (\lambda e^{-\lambda t_i}) = \sum_{i=1}^n (\ln \lambda - \lambda t_i)$$

$$\frac{\partial \ln \mathcal{L}(\lambda)}{\partial \lambda} = \sum_{i=1}^n \left(\frac{1}{\lambda} - t_i \right)$$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data
3. Write down likelihood function
4. Take log
5. Differentiate w.r.t. parameter of interest
6. Set to zero

$$f(t; \lambda) = \lambda e^{-\lambda t}$$

$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

$$\ln \mathcal{L}(\lambda) = \sum_{i=1}^n \ln (\lambda e^{-\lambda t_i}) = \sum_{i=1}^n (\ln \lambda - \lambda t_i)$$

$$\frac{\partial \ln \mathcal{L}(\lambda)}{\partial \lambda} = \sum_{i=1}^n \left(\frac{1}{\lambda} - t_i \right) = 0$$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data
3. Write down likelihood function
4. Take log
5. Differentiate w.r.t. parameter of interest
6. Set to zero
7. Find MLE for parameter of interest

$$f(t; \lambda) = \lambda e^{-\lambda t}$$

$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

$$\ln \mathcal{L}(\lambda) = \sum_{i=1}^n \ln (\lambda e^{-\lambda t_i}) = \sum_{i=1}^n (\ln \lambda - \lambda t_i)$$

$$\frac{\partial \ln \mathcal{L}(\lambda)}{\partial \lambda} = \sum_{i=1}^n \left(\frac{1}{\lambda} - t_i \right) = 0$$

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n t_i}$$

MLE for exponential rate parameter

1. Given exponential distribution
2. Given some data
3. Write down likelihood function
4. Take log
5. Differentiate w.r.t. parameter of interest
6. Set to zero
7. Find MLE for parameter of interest

$$f(t; \lambda) = \lambda e^{-\lambda t}$$

$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

$$\ln \mathcal{L}(\lambda) = \sum_{i=1}^n \ln (\lambda e^{-\lambda t_i}) = \sum_{i=1}^n (\ln \lambda - \lambda t_i)$$

$$\frac{\partial \ln \mathcal{L}(\lambda)}{\partial \lambda} = \sum_{i=1}^n \left(\frac{1}{\lambda} - t_i \right) = 0$$

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n t_i} = \hat{\tau}_{\text{MLE}}$$

heh

MLE of a function is the function of the MLE.

$$g = g(\theta)$$

$$\hat{g}_{\text{MLE}} = g(\hat{\theta}_{\text{MLE}})$$

$$\lambda = \frac{1}{\tau}$$

$$f(t; \lambda) = \lambda e^{-\lambda t}$$
$$\{t_1, t_2, \dots, t_n\}$$

$$\mathcal{L}(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i}$$

$$\ln \mathcal{L}(\lambda) = \sum_{i=1}^n \ln (\lambda e^{-\lambda t_i}) = \sum_{i=1}^n (\ln \lambda - \lambda t_i)$$

$$\frac{\partial \ln \mathcal{L}(\lambda)}{\partial \lambda} = \sum_{i=1}^n \left(\frac{1}{\lambda} - t_i \right) = 0$$

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n t_i} = \hat{\tau}_{\text{MLE}}$$

heh

Expectation of MLE for exponential rate parameter

$$E[\hat{\lambda}] = \quad ??$$

Expectation of MLE for exponential rate parameter

$$E[\hat{\lambda}] = \lambda \frac{n}{n-1} = \frac{1}{\tau} \frac{n}{n-1},$$

See bluebook Section 10.4.1 or work it out yourself

Expectation of MLE for exponential rate parameter

Biased estimator!

$$E[\hat{\lambda}] = \lambda \frac{n}{n-1} = \frac{1}{\tau} \frac{n}{n-1},$$

See bluebook Section 10.4.1 or work it out yourself

Expectation of MLE for exponential rate parameter

Biased estimator!

$$E[\hat{\lambda}] = \lambda \frac{n}{n-1} = \frac{1}{\tau} \frac{n}{n-1},$$

λ, τ related (reciprocal of one another)

But unbiased estimators don't necessarily remain unbiased after transformation

But MLEs are unbiased in large-sample limit

Circled term does go away as $n \rightarrow \infty$

MLE example: Gaussian

- 2 parameters
 - Mean: μ
 - Variance: σ^2

$$\mathbf{x} = \{x_1, x_2, \dots, x_n\}$$

$$\mathcal{L}(\theta) = P(\mathbf{x}|\theta)$$

$$= \prod_{i=1}^N P(x_i|\theta)$$

$$= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$\log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = \sum_{i=1}^n \left(\log \frac{1}{\sqrt{2\pi}} + \frac{1}{2} \log \frac{1}{\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right).$$

MLE example: Gaussian

$$\log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = \sum_{i=1}^n \left(\log \frac{1}{\sqrt{2\pi}} + \frac{1}{2} \log \frac{1}{\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right).$$

Take partial derivative w.r.t. μ , set derivative to zero, solve for μ

MLE example: Gaussian

$$\log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = \sum_{i=1}^n \left(\log \frac{1}{\sqrt{2\pi}} + \frac{1}{2} \log \frac{1}{\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right).$$

Take partial derivative w.r.t. μ , set derivative to zero, solve for μ

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Happens to be the sample mean

MLE example: Gaussian

$$\log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = \sum_{i=1}^n \left(\log \frac{1}{\sqrt{2\pi}} + \frac{1}{2} \log \frac{1}{\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right).$$

Take partial derivative w.r.t. μ , set derivative to zero, solve for μ

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

$$E[\hat{\mu}] = \mu$$

Unbiased

MLE example: Gaussian

$$\log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = \sum_{i=1}^n \left(\log \frac{1}{\sqrt{2\pi}} + \frac{1}{2} \log \frac{1}{\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right).$$

Take partial derivative w.r.t. σ , set derivative to zero, solve for σ

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

$$E[\hat{\mu}] = \mu$$

Unbiased

$$\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2.$$

Resembles the sample variance

But note the denominator..

MLE example: Gaussian

$$\log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = \sum_{i=1}^n \left(\log \frac{1}{\sqrt{2\pi}} + \frac{1}{2} \log \frac{1}{\sigma^2} - \frac{(x_i - \mu)^2}{2\sigma^2} \right).$$

Take partial derivative w.r.t. σ , set derivative to zero, solve for σ

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i.$$

$$E[\hat{\mu}] = \mu$$

Unbiased

$$\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2.$$

$$E[\widehat{\sigma^2}] = \frac{n-1}{n} \sigma^2.$$

Asymptotically unbiased

Estimators for Gaussian Variance

Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Unbiased

MLE

$$\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Biased but smaller

>

How to quantify errors on MLEs?

MLE is an estimator, depends on data

What is the spread of the MLE?

Variance on the MLE

Variance of estimator

1. Analytic method

- a. Stick expression for MLE into definition of variance and expand

$$\begin{aligned} V[\hat{\tau}] &= E[\hat{\tau}^2] - (E[\hat{\tau}])^2 && \text{Example using MLE for exponential } \tau \\ &= \int \dots \int \left(\frac{1}{n} \sum_{i=1}^n t_i \right)^2 \frac{1}{\tau} e^{-t_1/\tau} \dots \frac{1}{\tau} e^{-t_n/\tau} dt_1 \dots dt_n \\ &\quad - \left(\int \dots \int \left(\frac{1}{n} \sum_{i=1}^n t_i \right) \frac{1}{\tau} e^{-t_1/\tau} \dots \frac{1}{\tau} e^{-t_n/\tau} dt_1 \dots dt_n \right)^2 \\ &= \frac{\tau^2}{n}. \end{aligned}$$

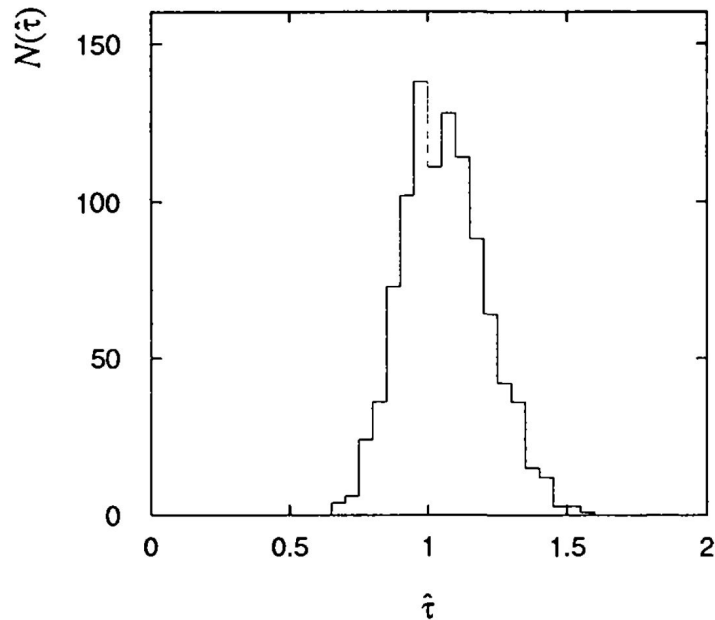
Variance of estimator

1. Analytic method

- a. Stick expression for MLE into definition of variance and expand

2. Monte-Carlo method

- a. If too difficult to solve analytically, integrate by MC



Variance of estimator

1. Analytic method

- a. Stick expression for MLE into definition of variance and expand

2. Monte-Carlo method

- a. If too difficult to solve analytically, integrate by MC

3. Cramer-Rao Bound method

- a. Too difficult to integrate by hand, too expensive to MC
- b. Only gives lower bound on variance

$$V[\hat{\theta}] \geq \left(1 + \frac{\partial b}{\partial \theta}\right)^2 / E \left[-\frac{\partial^2 \log L}{\partial \theta^2} \right]$$

Variance and Hessian

$$V[\hat{\theta}] \geq \left(1 + \frac{\partial b}{\partial \theta}\right)^2 / E \left[-\frac{\partial^2 \log L}{\partial \theta^2} \right]$$

Information inequality

$$(V^{-1})_{ij} = E \left[-\frac{\partial^2 \log L}{\partial \theta_i \partial \theta_j} \right].$$

Unbiased estimator

$$(\widehat{V^{-1}})_{ij} = -\frac{\partial^2 \log L}{\partial \theta_i \partial \theta_j} \Big|_{\theta=\hat{\theta}}.$$

Approximating using Hessian at MLE

$$\widehat{\sigma^2_{\hat{\theta}}} = \left(-1 / \frac{\partial^2 \log L}{\partial \theta^2} \right) \Big|_{\theta=\hat{\theta}}$$

Example

Variance of estimator

1. Analytic method

- a. Stick expression for MLE into definition of variance and expand

2. Monte-Carlo method

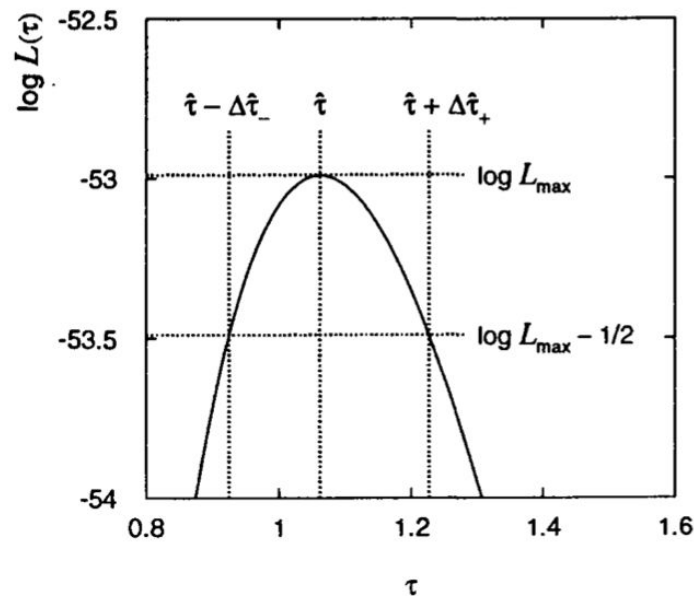
- a. If too difficult to solve analytically, integrate by MC

3. Cramer-Rao Bound method

- a. Too difficult to integrate by hand, too expensive to MC
- b. Only gives lower bound on variance

4. Likelihood method

- a. Max loglikelihood and drop 0.5
- b. Also gives 1σ central confidence interval on parameter



Variance from likelihood

Taylor expand around the MLE

$$\log L(\theta) = \log L(\hat{\theta}) + \left[\frac{\partial \log L}{\partial \theta} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2!} \left[\frac{\partial^2 \log L}{\partial \theta^2} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 + \dots$$

Variance from likelihood

Taylor expand around the MLE

$$\log L(\theta) = \log L(\hat{\theta}) + \left[\frac{\partial \log L}{\partial \theta} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2!} \left[\frac{\partial^2 \log L}{\partial \theta^2} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 + \dots$$

Very definition of MLE

$$\log L(\hat{\theta}) = \log L_{\max}$$

$$\widehat{\sigma^2}_{\hat{\theta}} = \left(-1 / \frac{\partial^2 \log L}{\partial \theta^2} \right) \Big|_{\theta=\hat{\theta}}$$

Variance from likelihood

Taylor expand around the MLE

$$\log L(\theta) = \log L(\hat{\theta}) + \left[\frac{\partial \log L}{\partial \theta} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2!} \left[\frac{\partial^2 \log L}{\partial \theta^2} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 + \dots$$

Very definition of MLE

$$\log L(\hat{\theta}) = \log L_{\max}$$

$$\log L(\theta) = \log L_{\max} - \frac{(\theta - \hat{\theta})^2}{2\widehat{\sigma^2_{\hat{\theta}}}},$$

$$\widehat{\sigma^2_{\hat{\theta}}} = \left(-1 / \frac{\partial^2 \log L}{\partial \theta^2} \right) \Big|_{\theta=\hat{\theta}}$$

Variance from likelihood

Taylor expand around the MLE

$$\log L(\theta) = \log L(\hat{\theta}) + \left[\frac{\partial \log L}{\partial \theta} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2!} \left[\frac{\partial^2 \log L}{\partial \theta^2} \right]_{\theta=\hat{\theta}} (\theta - \hat{\theta})^2 + \dots$$

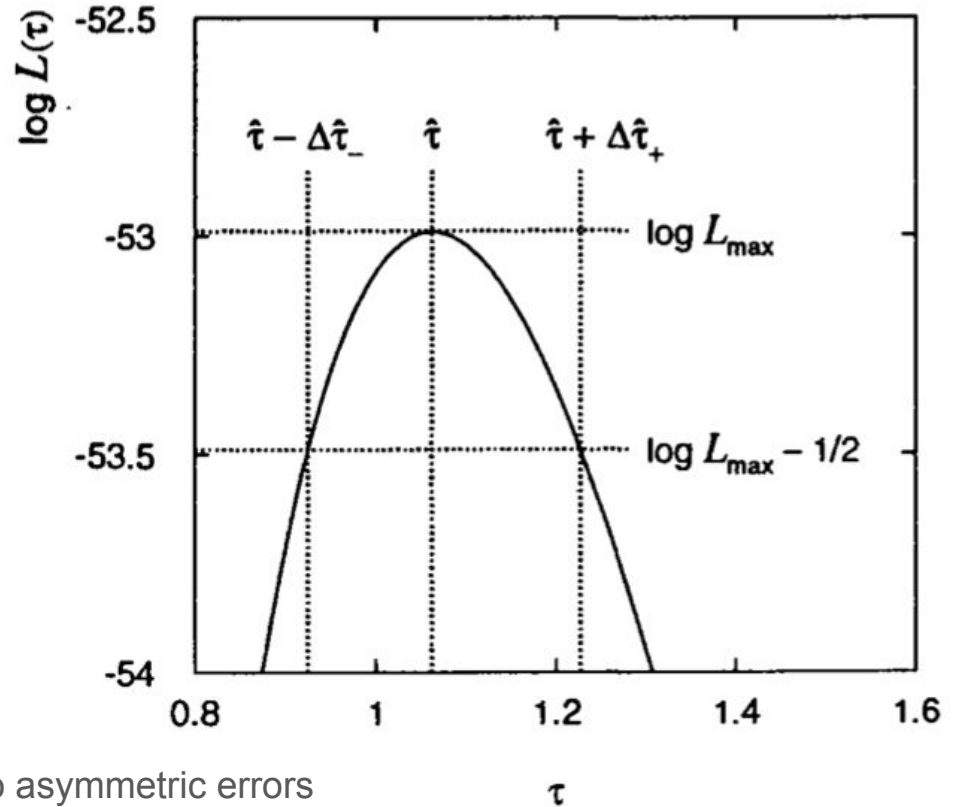
$$\log L(\theta) = \log L_{\max} - \frac{(\theta - \hat{\theta})^2}{2\widehat{\sigma^2_{\hat{\theta}}}},$$

$$\log L(\hat{\theta} \pm \hat{\sigma}_{\hat{\theta}}) = \log L_{\max} - \frac{1}{2}.$$

Variance from likelihood

$$\log L(\theta) = \log L_{\max} - \frac{(\theta - \hat{\theta})^2}{2\widehat{\sigma_{\hat{\theta}}}^2},$$

$$\log L(\hat{\theta} \pm \hat{\sigma}_{\hat{\theta}}) = \log L_{\max} - \frac{1}{2}.$$



Only approximation (Taylor expansion)

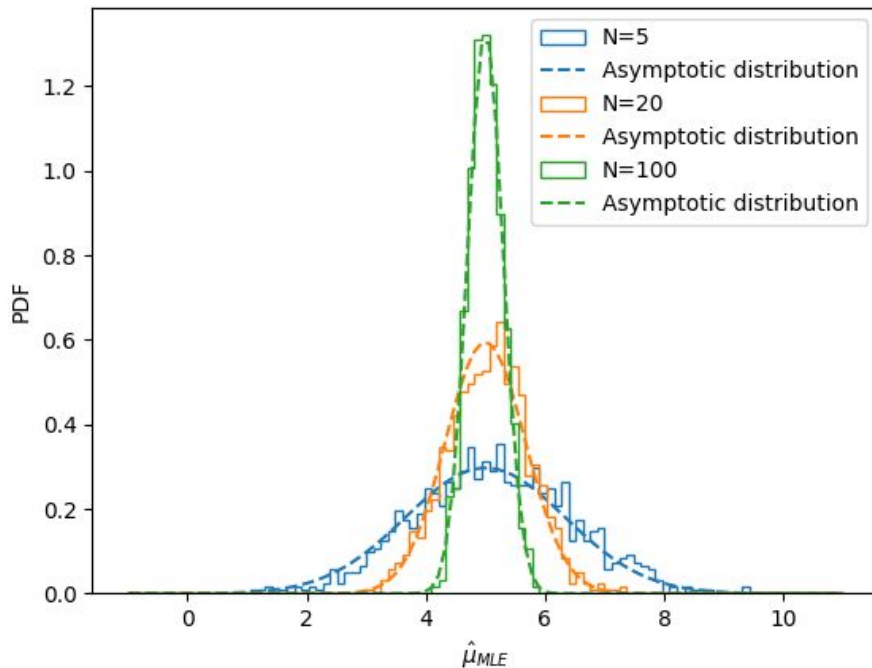
Proper method (interval computation) can give rise to asymmetric errors

Distribution of MLE → Gaussian in large-sample limit

- $\hat{\theta}_{\text{MLE}} \sim \mathcal{N}\left(E[\hat{\theta}_{\text{MLE}}], V[\hat{\theta}_{\text{MLE}}]\right)$
 - True for MLE for params for all distributions
 - E.g.: MLE for Gaussian expectation

$$\hat{\mu}_{\text{MLE}} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

- Stems from central limit theorem
 - See orangebook Section 7.3.1 for proof
- Good for sanity checks in simulations



Break

Extended Likelihood and code walkthrough
Back 11:40 am.

Extended Maximum Likelihood

Multiplication over the samples

$$L(\nu, \theta) = \underbrace{\frac{\nu^n}{n!} e^{-\nu}}_{\text{Poisson}} \underbrace{\prod_{i=1}^n f(x_i; \theta)}_{\substack{\text{Gaussian,} \\ \text{some} \\ \text{arbitrary} \\ \text{shape, etc}}} = \frac{e^{-\nu}}{n!} \prod_{i=1}^n \nu f(x_i; \theta).$$

Count

Shape

Likelihood can have many components

$$L(\nu, \boldsymbol{\theta}) = \frac{\nu^n}{n!} e^{-\nu} \prod_{i=1}^n f(x_i; \boldsymbol{\theta}) = \frac{e^{-\nu}}{n!} \prod_{i=1}^n \nu f(x_i; \boldsymbol{\theta}).$$


$$f(x; \boldsymbol{\theta}) = \sum_{i=1}^m \theta_i f_i(x),$$

Sum over the various components
Signal, background, etc

Example

$$f(x) = \frac{\mu_s}{\mu_s + \mu_b} f_s(x) + \frac{\mu_b}{\mu_s + \mu_b} f_b(x),$$

Show code time

Questions?

Thanks!

Backup slides

$$L(\nu, \boldsymbol{\theta}) = \frac{\nu^n}{n!} e^{-\nu} \prod_{i=1}^n f(x_i; \boldsymbol{\theta}) = \frac{e^{-\nu}}{n!} \prod_{i=1}^n \nu f(x_i; \boldsymbol{\theta}).$$

Take log

$$\log L(\boldsymbol{\theta}) = n \log \nu(\boldsymbol{\theta}) - \nu(\boldsymbol{\theta}) + \sum_{i=1}^n \log f(x_i; \boldsymbol{\theta})$$

$$= -\nu(\boldsymbol{\theta}) + \sum_{i=1}^n \log(\nu(\boldsymbol{\theta}) f(x_i; \boldsymbol{\theta})),$$

$$f(x; \boldsymbol{\theta}) = \sum_{i=1}^m \theta_i f_i(x),$$

$$\log L(\nu, \boldsymbol{\theta}) = -\nu + \sum_{i=1}^n \log \left(\sum_{j=1}^m \nu \theta_j f_j(x_i) \right).$$